GRANT & BUSINESS PLAN PORTFOLIO

Polymer DPP — NRW Grant Report

The Digital Product Passport infrastructure for Europe's polymer industry — business plan, business model, and grant-alignment documentation for NRW.BANK, EFRE.NRW, and regional circular-economy funding.

PREPARED FOR
NRW.BANK · EFRE.NRW · regional innovation grants

REGION
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01 — Executive Summary & Value Proposition

A verifiable trust asset, not another compliance spreadsheet

> 💡 **KEY TAKEAWAY**
>
> A working, deployed DPP platform — not a concept — that turns the EU's incoming ESPR/PPWR transparency mandate into a verifiable trust asset for NRW's polymer and chemical Mittelstand, built on cryptographic tamper-evidence and GS1-standard QR verification.

The problem

Europe's polymer value chain is being forced, in a short window, to prove things it has historically never had to prove: exactly how much recycled or bio-based content is in a given batch of resin, where that feedstock came from, what its carbon footprint is, and whether every claim on the label is independently verifiable rather than self-declared. The EU's Ecodesign for Sustainable Products Regulation (ESPR), the Packaging and Packaging Waste Regulation (PPWR, (EU) 2025/40), and the broader Digital Product Passport mandate turn "trust us" sustainability marketing into a hard compliance requirement with legal exposure attached — the EU's Green Claims direction explicitly targets unsubstantiated environmental claims.

For polymer compounders, recyclers, and converters — an industry segment heavily concentrated in NRW's Rhein-Ruhr and Rhineland chemical corridor — this creates an operational problem that spreadsheets and PDFs cannot solve at scale: mass-balance chain-of-custody (ISCC PLUS, REDcert2) has to be tracked batch-by-batch, certifications have to expire and get re-verified on a schedule, carbon footprint figures have to be distinguishable from marketing estimates, and every one of these facts has to be exposed through a QR-code-scannable, GS1-Digital-Link-compliant passport by the time ESPR's delegated acts start applying to plastics value chains.

Most mid-sized polymer processors — the Mittelstand backbone of NRW's chemical and plastics cluster — have neither the software engineering capacity nor the regulatory staff to build this themselves, and the generic ERP/PLM vendors serving this space were not built around DPP-specific data models.

What we have built

Polymer DPP is a working, deployed SaaS platform that gives polymer manufacturers, recyclers, and compounders a purpose-built Digital Product Passport system:

- **Industrial-grade material data capture** — ISO 1133 Melt Flow Index, density, tensile strength, PCR/PIR recycled content, bio-based content, and PEF/LCA-based carbon footprint, stored with exact decimal precision.
- **Multi-party trust workflow** — manufacturers submit batch data; independent, individually-credentialed third-party auditors formally verify or reject it before it is ever shown to a downstream buyer or consumer as fact.
- **Cryptographic immutability** — every batch record is SHA-256 sealed, and every state change is written to an append-only, hash-chained audit ledger with a live integrity-verification function.
- **Consumer- and auditor-facing transparency** — a GS1 Digital Link QR code on every batch resolves to a public passport page, plus dedicated auditor and buyer portals and a full ESG/compliance analytics suite.
- **Regulatory-adjacent analytics** — a computed Sustainability Readiness Index, PPWR 2030/2040 target tracking, supplier performance ranking, and illustrative EU plastic-tax exposure.

The value proposition

For manufacturers and recyclers: turn an unavoidable compliance burden into a competitive credential — a QR code any auditor can scan beats "we say our resin is 40% PCR."

For B2B buyers: a single point of truth for the sustainability data they must themselves report under their own CSRD/ESPR obligations, without re-keying supplier claims.

For NRW as a region: a homegrown, EU-compliant DPP infrastructure layer for one of Germany's largest chemical and plastics industrial clusters, reducing dependence on non-EU vendors for a function that becomes mandatory across the entire European polymer value chain within this decade.

Why now, why NRW

NRW is home to one of Europe's densest concentrations of chemical, plastics-processing, and recycling capacity, sitting directly alongside a world-class materials-science research base at RWTH Aachen. ESPR's delegated acts and PPWR's 2030 recycled-content thresholds create a hard deadline measured in a small number of years — the window to become NRW's default DPP infrastructure provider is open now.

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02 — Business Model Canvas

Value creation, revenue, cost, and market — at a glance

💡 **KEY TAKEAWAY**

Six B2B revenue lines anchored on a tiered manufacturer SaaS subscription, sold first into NRW's dense polymer/chemical cluster and expanded DACH/EU-wide as ESPR/PPWR compliance becomes mandatory everywhere.

VALUE PROPOSITIONS

- Verifiable — not self-declared — sustainability claims
- ESPR / PPWR / GS1 Digital Link compliance out of the box
- Industrial-grade, decimal-precise material data
- Cryptographic tamper-evidence
- One system for manufacturer → auditor → buyer trust

KEY PARTNERS

- RWTH Aachen / Fraunhofer research groups
- ISCC PLUS & REDcert2 certification bodies
- GS1 Germany (Digital Link standard)
- NRW chemical / plastics associations
- Neon (managed Postgres infrastructure)
- Independent auditors & auditing firms

KEY ACTIVITIES

- Platform engineering (schema, API, portal UX)
- Regulatory mapping as ESPR/PPWR evolve
- Auditor network onboarding & training
- Manufacturer onboarding & data migration

KEY RESOURCES

- The codebase (data model, ledger, QR engine)
- Embedded domain expertise (polymer science, EU compliance, security)
- Live production database & verified batch history
- Auditor & manufacturer pilot relationships

CUSTOMER RELATIONSHIPS

- Direct B2B sales with onboarding support
- Self-service supplier-invite links
- White-glove auditor onboarding
- Long-term account relationships

CHANNELS

- Direct sales into the NRW chemical corridor
- Industry association & cluster events
- Certification-body co-marketing
- The public QR-scan page as a brand touchpoint

CUSTOMER SEGMENTS

- **Primary:** mid-sized compounders & recyclers (50–500 employees), NRW/DACH
- **Secondary:** B2B buyers (OEMs, converters) needing verified material data
- **Tertiary:** independent auditors & certification-adjacent providers

COST STRUCTURE

- Engineering & product development (dominant, early-stage)
- Cloud infrastructure (Neon, hosting)
- Compliance & domain expertise (ongoing)
- Sales, partnerships & customer success
- Security/audit & certification maintenance
- Grant & compliance reporting overhead

REVENUE STREAMS

- ① Tiered manufacturer subscription (core)
- ② Auditor / verification seat licensing
- ③ B2B buyer / API access
- ④ Compliance export & certificate services
- ⑤ Onboarding, migration & integration services
- ⑥ Future: white-label / multi-tenant licensing

Revenue streams — the detail

The platform is designed around recurring, usage-aligned B2B SaaS revenue rather than one-off project fees, reflecting that a Digital Product Passport only has value if it stays live and current for the lifetime of every batch it describes.

- **Tiered manufacturer subscription** — the core revenue line, tiered by batch volume and feature access (base: passport creation, QR/GS1, public page; higher: genealogy, carbon calculator, ESG/PPWR analytics, green-tax estimation, bulk certificate/label generation).
- **Auditor/verification seat licensing** — auditors are a distinct, professionally liable user class, individually credentialed and priced separately.
- **B2B buyer/API access** — a stable, versioned Buyer API for OEMs and converters feeding verified supplier data into their own ERP/CSRD reporting, priced per connected account or data volume.
- **Compliance export & certificate services** — usage-based fees for PDF Certificates of Conformity, printable QR label sheets, and audit-ready compliance export workbooks with an embedded live hash-chain integrity check.
- **Onboarding, migration & integration services** — one-time professional-services fees for larger accounts, priced separately from recurring SaaS revenue.
- **(Future) White-label / multi-tenant licensing** — a plausible higher-value, lower-volume line once the multi-role architecture is proven at scale — flagged as a post-MVP option, not a Year 1 assumption.

Cost structure — the detail

Engineering and product development dominate the early-stage cost line, alongside cloud infrastructure (structurally low per-passport cost thanks to a stateless, compute-light architecture), ongoing compliance and domain expertise (a recurring cost, not a one-time setup, as ESPR/PPWR delegated acts continue to be published), sales and customer success (churn is existential here — a customer's entire batch history and public QR resolvers depend on continuity of service), periodic security/audit review, and the reporting overhead that comes with public co-financing.

Target market: NRW and EU

Immediate: NRW's polymer and chemical processing base — the Rhineland chemical corridor and Ruhr-area recycling sector — targeting compounders, recyclers, and converters (50–500 employees) already producing PCR/PIR or bio-based grades. **Adjacent:** because ESPR and PPWR are EU regulations, every polymer manufacturer across the EU faces the same deadline — NRW is the beachhead (home region, RWTH Aachen access, dense customer concentration), with a clear expansion path into DACH and then Benelux/broader EU.

03 — NRW Grant Alignment

Where this project fits NRW's own strategic priorities

⚠ **READ BEFORE FILING ANYTHING**

Program names, ceilings, co-financing rates, and deadlines change between calls. Every program below fit this project's profile as of August 2026 research — confirm current terms on the linked page immediately before submitting.

Strategic fit

NRW's state government and NRW.BANK organize funding around a small number of standing priorities this project maps onto directly: **sustainability & circular economy** (Ressource.NRW's resource-efficiency line matches PCR/PIR tracking and mass-balance chain-of-custody exactly); **digitalization of traditional Mittelstand** (MID exists specifically to fund digitalization of SME compliance and supply-chain processes); **EU regulatory readiness for regional industry** (the whole NRW chemical sector faces the ESPR/PPWR deadline together, the explicit logic behind EFRE.NRW/JTF's "Gründung und Digitalisierung" priority); and **research-to-market translation** via the state's own science base — a criterion this project's RWTH Aachen fit answers directly, not decoratively.

Candidate NRW / EU funding instruments

All require, at minimum, a registered establishment (Betriebsstätte) in NRW and fall within defined SME (KMU) size thresholds for the most favorable terms.

DIGITAL + SUSTAINABLE

NRW.BANK.Invest Zukunft

A financing/investment product explicitly targeted at transforming businesses digitally and sustainably — the most direct match for a SaaS platform whose entire value proposition is exactly that transformation.

nrwbank.de ↗ 2026 program page ↗

SME DIGITALIZATION

Mittelstand Innovativ & Digital (MID)

NRW.BANK's core SME digitalization line, incl. a voucher sub-track. Relevant both for this company and as a co-financing angle for prospective manufacturer customers adopting the platform.

MID overview ↗ Digitalisierung Gutscheine ↗

EU CO-FINANCED

EFRE/JTF-Programm NRW 2021–2027

NRW's implementation of the EU's ERDF/EFRE and Just Transition Fund, with an explicit "Gründung und Digitalisierung" priority — the natural home for a growth-stage digital startup building regionally significant infrastructure.

Program page ↗ efre.nrw ↗

CIRCULAR ECONOMY

Ressource.NRW (2025–2026)

A dedicated resource-efficiency/circular-economy program run via EFA.NRW — the most directly on-theme application for a platform whose core function is PCR/PIR tracking and material-flow transparency.

efa.nrw ↗

ADVISORY / FIRST STOP

Innovationsförderagentur NRW (IN.NRW)

The state's central coordination point for innovation funding — useful less as a single program and more as a first-contact advisory channel to identify the best-fitting current call.

in.nrw ↗

OVERVIEW

NRW.BANK — SME funding overview

A consolidated overview of NRW.BANK's SME products for innovation, digitalization, and know-how transfer — a good cross-check for newer or better-fitting programs.

nrwbank.de ↗

RWTH Aachen ecosystem synergy

RWTH Aachen's **Center for Circular Economy (CCE)** is a standing, well-resourced research center organized around exactly the domain this platform digitizes. (cce.rwth-aachen.de ↗)

① Technical validation

An advisory relationship with CCE (or an adjacent RWTH group) gives grant reviewers independent confirmation that the platform's emission-factor methodology and material-data schema reflect current materials-science practice.

② Industry credibility by association

RWTH Aachen already partners with major chemical/materials companies on circular-economy research (e.g. Covestro), showing the region's flagship industrial players already treat this ecosystem as a serious partner.

③ Talent & future R&D pipeline

Proximity to RWTH's materials-science graduate programs is a practical talent argument, and a plausible basis for a future joint research application building on the platform's accumulated verified batch data.

✅ **RECOMMENDATION**

Before filing any application that cites this synergy, establish at minimum an informal advisory or letter-of-support relationship with a named RWTH Aachen contact — reviewers weight a named, real collaboration far higher than a general regional-proximity claim.

Positioning summary for the application narrative

> *"Polymer DPP builds the digital infrastructure NRW's chemical and plastics Mittelstand needs to meet a legally mandated EU transparency requirement that is already approaching, turning a compliance cost into an auditable trust asset — using a technical architecture developed in NRW and validated against the region's own materials-science research base."*

Adapt this framing to each program's own priority language (digitalization vs. circular economy vs. regional structural development) rather than submitting identical text to each — reviewers score against different published criteria.

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04 — Technical Architecture Summary

What's actually live — written for a grant reviewer

> 💡 **KEY TAKEAWAY**
>
> Every capability below is live in the production codebase today — a cryptographic hash-chained audit ledger, GS1 Digital Link QR, a multi-role RBAC trust workflow, and PPWR-aware ESG analytics — not a roadmap promise.

Next.js (App Router) | TypeScript | PostgreSQL — Neon | Prisma ORM | Prisma.Decimal precision | SHA-256 hash-chained ledger | GS1 Digital Link | @react-pdf/renderer

A grant reviewer evaluating a digitalization or circular-economy project typically needs to answer two questions beyond "does the idea make sense": is there a real, working system behind this application, and is the technical approach sound enough to scale and survive regulatory/security scrutiny. This section answers both directly — the platform is not a prototype, it is a deployed application with a live PostgreSQL database, real API routes, and multiple working user-facing portals.

The Digital Product Passport data model

At the center of the schema is `PolymerPassport` — one row per physical batch — holding GTIN, batch ID, polymer type and grade, ISO 1133 melt flow index, density, tensile strength, recycled content (PCR/PIR distinction), bio-based content, and a verified carbon footprint figure, optionally third-party-verified. Every passport carries a SHA-256 `recordHash` and, once revised, a `previousRecordHash`, so the record's own history is independently checkable.

Built on top of this core, already live: **batch genealogy** (a recursive upstream traceability tree, capped at a bounded depth for query safety); a **carbon footprint calculator** kept strictly separate from the manufacturer's own verified LCA figure and always labeled illustrative; a **certification workflow** (ISCC PLUS, ISO 14021, EN 15343, REDcert2) with a formal pending-review → verified/rejected lifecycle, with expiry computed at read time rather than stored as a manually maintained flag; and a real **GS1 Digital Link QR** resolver with a printable label-sheet suite for factory-line batch labeling.

Cryptographic immutability and the audit ledger

Every consequential state change is written to an append-only `AuditLog` inside the same database transaction as the change itself. Each entry is SHA-256 hashed over its own fields plus the previous entry's hash, forming one global, insertion-ordered hash chain — the same structural principle blockchains use for tamper-evidence, without distributed-ledger overhead. Two independent layers of protection follow: a live integrity-verification function that mathematically proves whether any row has been altered, and a database-level append-only trigger that rejects any `UPDATE`/`DELETE` outright. A dedicated Audit Log interface lets an administrator filter, page, and export the full chronological trail, including a compliance-export workbook that stamps a live integrity-check result into the file at export time.

Role-based access and multi-party trust

A genuine multi-party trust model, not a single-tenant admin tool: **manufacturers** create and manage batch passports; **independent auditors** each hold their own individually issued cryptographic credential — not a shared role key, because a formal sign-off is only meaningful if attributable to one accountable person — and review submissions through a dedicated Auditor Portal; **B2B buyers** access their purchased-batch portfolio through a versioned Buyer API and read-only Customer Portal; **upstream suppliers** can be invited via a token-based, self-service submission flow without requiring full accounts; and **consumers** scanning the public QR code see only verified, publicly appropriate data — enforced server-side, not just in the UI.

ESG, compliance, and analytics layer

Computed fresh on every request rather than cached, so it can never disagree with the underlying records: PCR/PIR mass tracking and illustrative EU plastic-tax exposure; a certification compliance breakdown with live-computed expiry alerting; a transparent, weighted Sustainability Readiness Index; PPWR 2030/2040 recycled-content target tracking benchmarked against the regulation's published Article 7 thresholds; and a supplier performance/compliance ranking dashboard. Every computed score shows its full component breakdown and weighting rather than a bare number.

🛠️ **PLANNED, NOT YET BUILT — STATED FOR TRANSPARENCY**

Production-grade session authentication for the manufacturer/admin console (current interim mechanism uses shared per-role API keys, adequate for a pilot); a first-class batch-weight field so mass-based ESG metrics cover every batch, not only those where the carbon calculator was run; and formal third-party security/penetration review ahead of handling external paying-customer data at scale.

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05 — Roadmap & Financial Projections

From pilot validation to DACH/EU expansion

⚠️ **ALL FIGURES BELOW ARE ILLUSTRATIVE PLACEHOLDERS**

Sized to a plausible NRW polymer-compliance SaaS trajectory — not audited, not based on signed contracts. Replace with the team's own bottom-up model before any binding submission.

Phase 0 — Foundation COMPLETE

Already delivered

- Core platform live: passport data model, cryptographic audit ledger, GS1 QR/Digital Link engine, certification workflow, auditor & buyer portals, ESG/compliance analytics, supplier onboarding, multi-language public page.

Phase 1 — Pilot validation NEXT

Months 1–6

- Onboard 3–5 pilot manufacturer customers across compounding, recycling, and converting
- Onboard a small initial auditor network (2–4 auditors or one firm)
- Formalize the RWTH Aachen / CCE advisory relationship
- Harden production-grade session auth; complete first external security review
- File the first NRW.BANK / EFRE.NRW / Ressource.NRW application(s)

Phase 2 — Regional scale-up

Months 7–18

- Convert pilots to paying subscriptions; expand sales across the NRW Mittelstand base
- Ship the Buyer API/CSRD-integration tier commercially with an anchor OEM/converter
- Complete first-class batch-weight field & full mass-based ESG coverage
- Formalize the auditor network into a structured, trained program

Phase 3 — DACH/EU expansion

Months 19–36

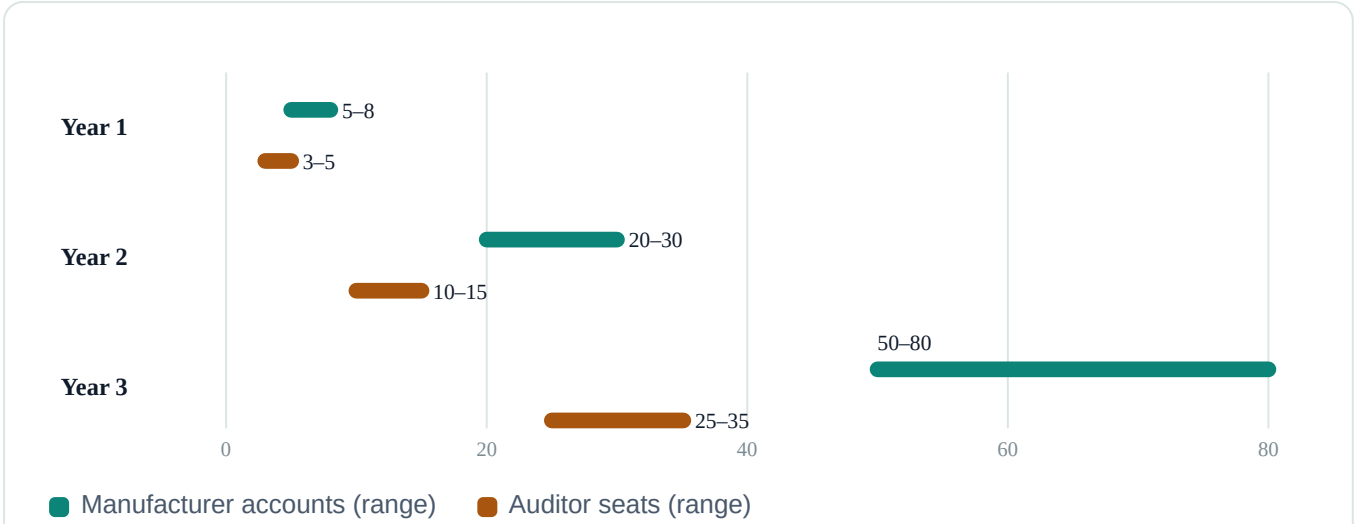
- Expand into the wider DACH chemical cluster on NRW reference customers
- Evaluate white-label/multi-tenant licensing as a second growth vector
- Pursue a follow-on research/translation grant on the platform's verified dataset
- Reassess entry into Benelux/broader EU as ESPR delegated acts take effect there

Milestone summary

MILESTONE	TARGET	SUCCESS SIGNAL
First 3–5 pilot manufacturer customers live	Month 3–6	Real production batches issued, not demo data
Session auth + external security review complete	Month 6	No critical findings outstanding
First paying subscription conversions	Month 6–9	≥1 pilot converts to paid
Buyer API live with ≥1 anchor buyer	Month 12–15	A real OEM/converter consuming verified data
Auditor network formalized (≥5 active)	Month 12–18	Verification turnaround SLA published
NRW.BANK/EFRE.NRW-type grant secured	Month 6–12	Co-financing awarded, reporting in place
DACH expansion sales motion launched	Month 18–24	First non-NRW paying customer

Illustrative financial projections

A simple, transparent structure: a per-manufacturer-account monthly subscription (blended across tiers), a smaller per-seat auditor line, and a services/API line that grows as the Buyer API matures.



Illustrative ranges only — not audited, not based on signed contracts. See the milestone table above for the conditions each range assumes.

	YEAR 1	YEAR 2	YEAR 3
Paying manufacturer accounts (year-end)	5–8	20–30	50–80
Auditor seats (year-end)	3–5	10–15	25–35
Buyer API / services revenue	Pilot only	Growing	Established line
Primary cost drivers	Engineering, domain expertise, infra	+ sales, customer success, security review	+ partnerships, DACH expansion costs
Grant co-financing role	Bridges pilot-phase cost gap	Supports scale-up & security hardening	Supports expansion-phase R&D

Modeling notes for the team — replace before submission

- Anchor subscription tiers to actual pilot pricing conversations, not a top-down guess — B2B pricing here is usually set by the customer's cost of *non-compliance*, not a generic SaaS benchmark.
- Build the cost-structure line items into an actual monthly burn model before stating any runway or grant-need figure — NRW.BANK/EFRE.NRW applications typically require a documented use-of-funds breakdown.
- State a specific funding ask only after checking the current ceiling and co-financing rate of the target program — do not carry a number from this template into a real application.

- Where a program requires job-creation or regional-impact projections, derive those from the headcount line rather than a separate, disconnected number.

Before you file

Submission checklist & caveats

Four things to resolve before any of this becomes a binding application, carried over from the source package's own README.

1. **Verify current program parameters before filing**
   Funding ceilings, co-financing rates, deadlines, and eligibility rules for NRW.BANK and EFRE.NRW programs change between calls. Re-check the live program pages immediately before submitting — the figures here were not locked in.

2. **Financial projections are illustrative, not audited**
   Section 5's figures must be replaced with the founding team's own bottom-up model (headcount costs, actual pilot pricing, real pipeline) before being submitted as binding numbers.

3. **No claims about certifications the platform doesn't hold**
   The platform supports ISCC PLUS / ISO 14021 / EN 15343 *data capture and verification workflow* — it is not itself a certification body. Never let grant text imply accreditation the product doesn't have.

4. **Legal entity and eligibility details are placeholders**
   Company legal form, registered NRW address, headcount, and SME (KMU) status are the applicant's own facts to fill in — most programs restrict eligibility to a registered NRW establishment and defined SME size classes.

Polymer DPP — NRW Grant & Business Plan Portfolio · Compiled 2026-09-18 from the six-document `docs/grant-nrw/` package · Prepared by Reza Farzad (reza.h.farzad@gmail.com) · Working draft — not audited or legally reviewed. Source Markdown files live alongside this report for editing and version control.